



جامعة السلطان قابوس
Sultan Qaboos University

Project Title : Official Website Of Car Dealership

Course : COMP3700 (introduction to Web Computing)

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Project Overview And Objectives:

Overview:

This project involves creating an Official Car Dealership Website to modernize the car-buying experience. Instead of visiting physical showrooms, customers can browse a full inventory, check prices, and view car specifications from home. The site aims to save time and reduce the effort of searching for a vehicle by providing a "digital showroom" that is accessible 24/7.

Main objectives:

- **27/7 Access:** Allow customers to browse cars anytime and anywhere.
- **Smart Filtering:** Enable users to search by price, brand, car type and more.
- **Detailed Specs:** Provide full technical details and features for every car.
- **Price Clarity:** Show clear pricing to help users find the car matches their budget.
- **Easy Booking:** Let users schedule test drives or contact the dealer easily.
- **Wider Reach:** increase the dealership's visibility to online shoppers across the country.

Target Audience and stakeholders:

Target Audience groups :

- Individual car buyers such as Students, Families, wealthy persons, Employees and etc.

Stakeholders and Their Needs:

- **Dealership Owner** : Smart and effective site to increase sales.
- **Sales team**: easy way to manage car listing and receive customer inquiries.
- **Web admins**: secure, fast, platform that work for all devices.

Stakeholder interview :

- **Person 1 :**
 - **Need** : high quality 360 photos and clear pricing without having call.
 - **Requirement**: The site must have price filter and compare tool.
- **Person 2 :**
 - **Need** : Want to Know which car customers are more interested.
 - **Requirement**: The page must have ability to track users most search cars and “Book test drive” that send car details to sales man.

Similar Websites:

Similar Websites :

- **Chevrolet official website** : <https://www.chevrolet.com/>
- **Yalla Motor** : <https://oman.yallamotor.com/auto-companies>

Similar Websites	How My Website is Different	How my Website will be Better
 Chevrolet	This Website Have limited types of cars (only Chevrolet cars)	My website will provide different brands and types of cars and test drive booking.
 Yalla Motor	This portal is filled with annoying ads and have outdated listing.	Will be easy to use and focus of car shopping with no ads and provide

		direct searching for actual available cars
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Website Features And Functionalities:

- **Advanced Vehicle Search and Filtering:** Users can filter cars by make, model, year, price range, and fuel type to quickly find vehicles that meet their specific needs.
- **Detailed Vehicle Listing Pages:** Each car will have a dedicated page featuring high-resolution images, comprehensive technical specifications, and history reports to ensure transparency for buyers.
- **Interactive Loan Calculator:** A tool allowing potential buyers to estimate monthly payments based on interest rates and down payments, aiding in financial planning.
- **User Profiles and Favorites:** Customers can create accounts to save their favorite listings and track their inquiries, providing a personalized browsing experience.
- **Online Test Drive Booking:** An integrated calendar system where users can schedule appointments with sellers or dealerships directly through the platform.
- **Seller Dashboard:** A dedicated interface for sellers to upload vehicle details, manage active listings, and communicate with interested buyers.

Website Logo Design:



Logo Explanation:

- **Shapes:** The logo features a sleek, stylized silhouette of a car integrated with a circular "gauge" or "steering wheel" element to represent motion and the automotive industry.
- **Colors:** A palette of Metallic Silver and Deep Navy Blue is used. Silver conveys modern technology and high-quality machinery, while Navy Blue represents trust, professionalism, and reliability—essential for high-value transactions like car sales.
- **Typography:** A bold, sans-serif font is utilized to demonstrate strength and clarity, ensuring the brand name is easily readable across different digital devices.
- **Theme:** The minimalist design aligns with a modern, efficient marketplace identity, emphasizing a seamless car-buying journey.

Website Content And Page Descriptions:

In addition to the standard "index," "about us," and "contact us" pages, the website will include the following five pages:

- **Inventory/Browse Cars:** This page displays all available listings. It serves the main objective of connecting buyers with products through a clean, grid-based layout and search tools.
- **Sell Your Car:** A guided submission page where owners can enter vehicle details (VIN, mileage, condition) and upload photos. This serves the objective of expanding the site's database.
- **Financing & Insurance:** Provides educational content regarding car loans and links to the payment calculator. This page serves the objective of assisting users with the financial aspects of a purchase.
- **Vehicle Comparison:** A specialized tool page where users can select up to three cars to view their specifications side-by-side, serving the objective of informed decision-making.
- **User Dashboard/Account:** A private area for logged-in users to manage their profiles, saved cars, and messages, serving the objective of user engagement and retention.

Database Design And Data Management:

Data Types:

1. **User Data:** Names, Passwords, Phone number and Record interactions (such as favorite cars).
2. **Vehicle Data:** Brand, Model, Year, Price, Condition (New/Used), and Technical Specifications
3. **Operational Data:** Test drive bookings, financing request, messages and buying request.

Data Exchange:

- **From customer to management:** When a customer fills out a "Finance Request" or "Appointment Booking" form, the data (name, income, vehicle type) is sent to the staff dashboard.
- **From management to customer:** The employee updates the order status (accepted/rejected), and the result appears in the customer's control panel or via email notification.

- **Inventory updates:** Once a sale is completed, the vehicle's status in the database is updated so that it immediately disappears from the public display for visitors.

Security Measures:

1. **Password encryption:** Use algorithms such as BCrypt or Argon2; the password must never be stored as plain text.
2. **Protecting data in transit:** Enable the HTTPS (SSL/TLS) protocol to encrypt data exchanged between the browser and the server.
3. **To verify permissions (RBAC):** Use a "roles-based access control" system so that the customer cannot access another customer's data or modify car prices.
4. **Preventing SQL Injection Attacks:** Use prepared statements to ensure that data entered into search forms does not harm the database.

Web Hosting Selection And Justification:

We saw one of (**Ooredoo Data2Cloud / Zayn Technology**) is the best hosting and we choose **Zayn Technology**.

Justification for Choosing Zayn Technology (Shared Hosting):

1. The balance between cost and effectiveness: Zayn Technology offers affordable packages in Omani Rial suitable for small database sizes as a starting point.
2. Since the exhibition targets customers within the Sultanate of Oman, hosting the website on Zayn's locally located servers ensures extremely high loading speeds. This geographical proximity reduces latency.
3. Full support for the required technologies (Technical Compatibility):
The hosting service offers 100% compatibility with the project's technical requirements:
 - It supports PHP and MySQL for managing the vehicle and customer database.
 - It provides a free SSL certificate, essential for protecting user data and securing logins.
 - It features an easy-to-use cPanel control panel for uploading XHTML, CSS, and JavaScript files.

4. Reliability and Local Support: Zayn Technology offers local technical support, ensuring easy communication to resolve any urgent technical issues within the same time zone. It also guarantees an uptime rate of up to 99.9%, which is essential to ensuring customers have access to stock at any time.
5. Scalability: Although it is shared hosting, the provider allows for easy upgrades. If the showroom expands and the number of cars and visitors increases significantly, we can seamlessly migrate to a virtual private server (VPS) with the same provider without having to rebuild the website from scratch.

Website Prototype (Google Sites + Barcode) :

<https://sites.google.com/student.squ.edu.om/carhome/home>



Ethical, Social & Professional Considerations:

Societal Impacts:

1. **Digital Transparency:** The website contributes to changing the traditional car-buying culture in society by providing transparent maintenance records, publicly available prices, and accurate technical comparisons. This reduces opportunities for manipulation or "commercial fraud" and increases consumer awareness before making a major financial decision.
2. **Support for Digital Transformation:** The project aligns with Oman Vision 2040 by reducing the need for physical travel between exhibitions, thus reducing traffic congestion and carbon emissions, and saving time and effort for citizens and residents by completing transactions (such as booking a test drive) electronically.

Ethical & Professional Challenges:

1. **Data Accuracy & Integrity:** Present the car's true condition (such as previous accidents or mechanical defects). Professionally, the buyer should not be misled by altered photos or incomplete information.
2. **Data Privacy:** The website collects sensitive information such as phone numbers, identities, and financial data for funding applications. The challenge is preventing this data from being leaked to marketing companies or third parties without permission.
3. **Algorithmic Fairness:** When arranging the appearance of cars or recommending specific cars to the user, there should be no bias towards certain cars due to a "higher commission" but rather based on the user's actual needs.

Action Plan:

1. **Verification System:** The website will include a special section for independent vehicle inspection reports (such as computer scans), with each vehicle linked to its chassis number (VIN) to ensure credibility.
2. **Strict Privacy Policy:** Implementing a comprehensive end-to-end encryption system for forms containing personal data, with a clear document (Terms & Conditions) agreed upon by the user explaining how their data will be used.
3. **Full Disclosure:** Make a mandatory field should be placed in the database for each car indicating "accident history" or "known defects," so that the advertisement cannot be published without filling it in, thus ensuring professional integrity.

Updates to The project (project part 4) :

responsibilities assigned to each team member :

Name :

Task:

Alwarith Saud Al Hadhrami	JavaScript Form Validation
Ahmed Fahad Al Wahaibi	Database Design and Implementation
Mohammed Khalid Al Kindi	Form Processing and Data Retrieval with PHP
Moayad Nabbel Al Jabri	Data Access Using PHP Web Pages
All team Members	Team Collaboration, GitHub, and Report

A description of our database design :

The database is designed to support a vehicle marketplace and feedback system. It utilizes a relational structure to manage inventory, user interactions, and administrative feedback.

1. Core Tables & Schema

The design is centered around two primary functional areas: Inventory Management and User Feedback.

- Cars Table

This table acts as the central repository for all vehicle listings. It stores the technical and descriptive attributes required for the "Cars Browse" interface.

- **Attributes:** Includes unique identifiers for each vehicle, make, model, year, price, and status.

- **Functionality:** Supports the [`delete_car.php`] logic by allowing specific records to be removed from the inventory via their unique ID.

- Feedback Table

This table handles the social and quality-control aspect of the platform, as seen in the "Feedback Browse" interface.

- **Attributes:** Stores user-submitted comments, ratings, and timestamps.
- **Relational Links:** It connects user entries to specific transactions or general site experiences.

2. User Interface Integration

The database design directly dictates the layout of the front-end components:

- **Cars Browse:** The query logic fetches data from the Cars table to populate the grid view of available vehicles.
- **Feedback Browse:** This view aggregates data from the [`Feedback`] table, displaying community reviews and administrative responses.

3. Data Integrity & Operations

The SQL implementation includes constraints to ensure data consistency:

- **Primary Keys:** Every car and feedback entry is uniquely indexed to prevent duplicate records.
- **Operational Scripts:** The [`delete_car.php`] script facilitates secure communication with the database to execute [`DELETE`] queries based on administrative actions taken in the UI.

Detailed explanation of our PHP code and its interaction with the database :

1. Database Design and Implementation

This section describes the backend structure that powers the AutoTrade system.

Database Schema Overview

The system utilizes a relational database named [`autotrade_database`] to manage inventory and user interactions. It consists of two primary tables:

- **Cars Table:** This table serves as the inventory repository, storing essential data for each vehicle including its unique ID, make, model, year, and price.
- **Feedback Table:** This table captures user-submitted data, allowing the administration to monitor customer reviews and ratings.

Data Integrity

- **Primary Keys:** Unique identifiers are assigned to each record in the [`Cars`] and [`Feedback`] tables to ensure data accuracy and prevent duplication.
 - **Relational Logic:** The structure is optimized to allow seamless retrieval of vehicle details for the "Cars Browse" interface.
-

2. PHP System Functionality

This section explains how the PHP scripts serve as the engine for the website, connecting the user interface to the database.

Database Connectivity

The system uses a centralized connection script (`db_connection.php`) to establish a secure link between the web server and the MySQL database. This script provides the necessary credentials to authorize data transactions.

Dynamic Data Presentation

The PHP code enables dynamic content generation for the following pages:

- **Cars Browse:** The script executes a [`SELECT`] query to fetch all available vehicles from the database and displays them in a structured grid for the user.
- **Feedback Browse:** This module retrieves community ratings and comments, allowing users to see historical feedback stored in the system.

Administrative Operations

The system includes administrative tools for inventory management:

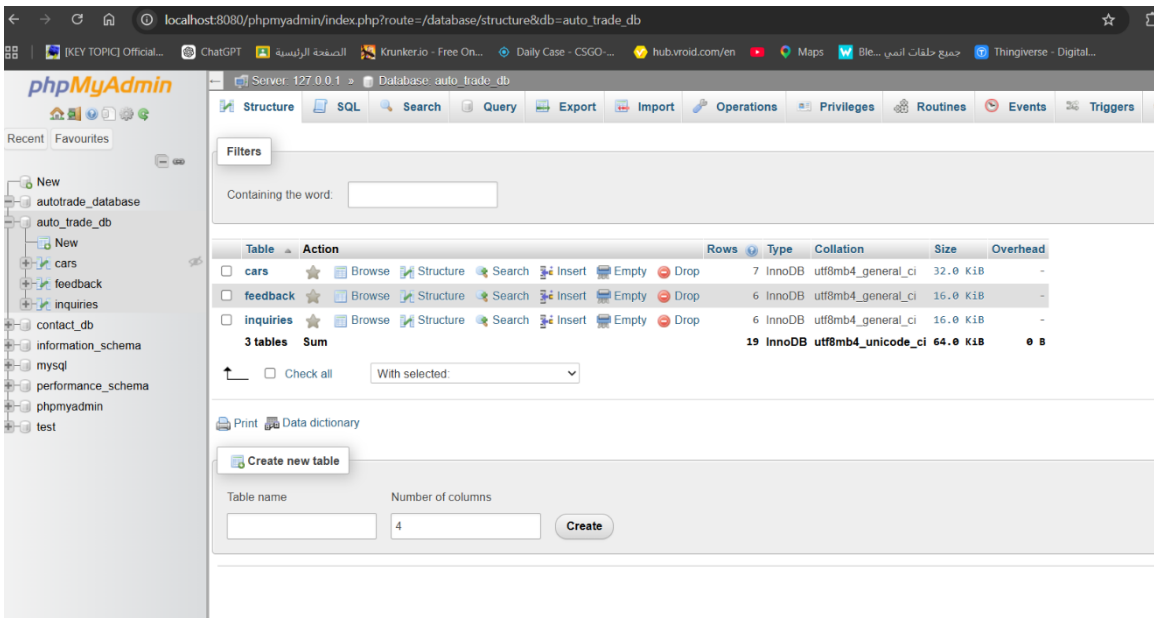
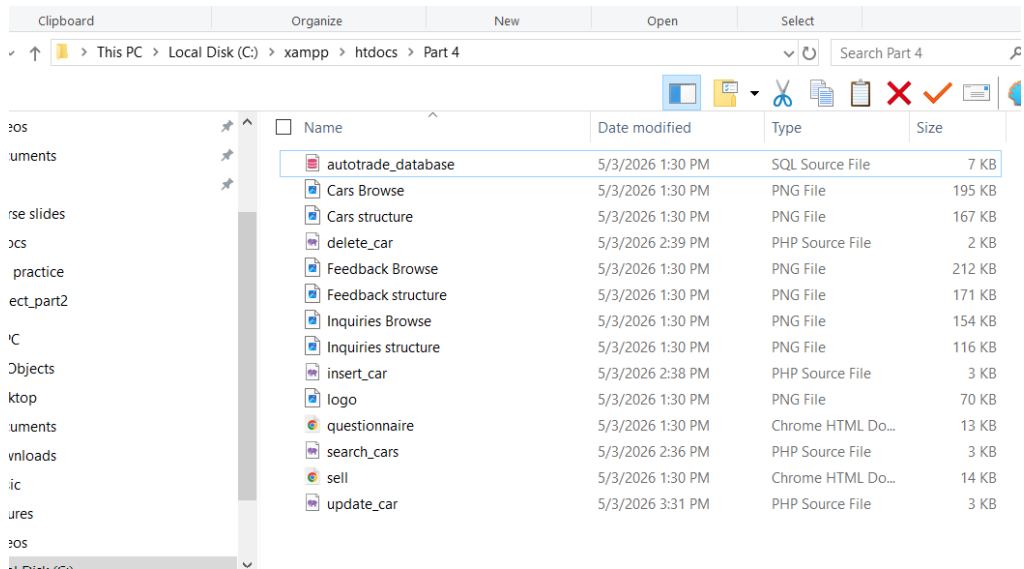
- Delete Car Logic: Through the [`delete_car.php`] script, the administrator can remove specific listings. The script identifies the vehicle by its ID and executes a [`DELETE`] SQL command to update the database in real-time.

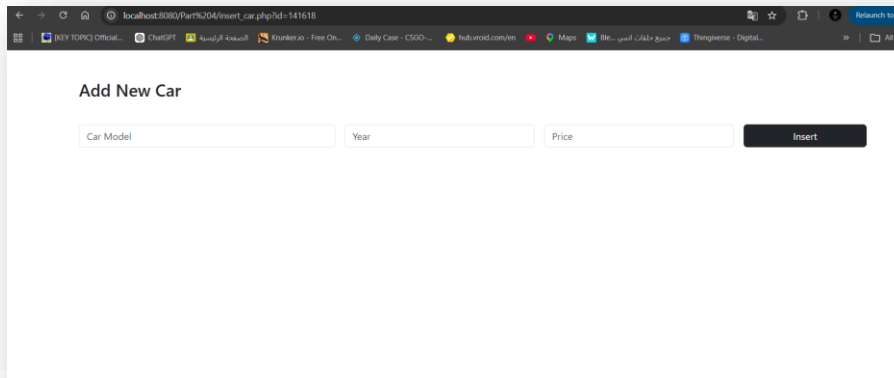
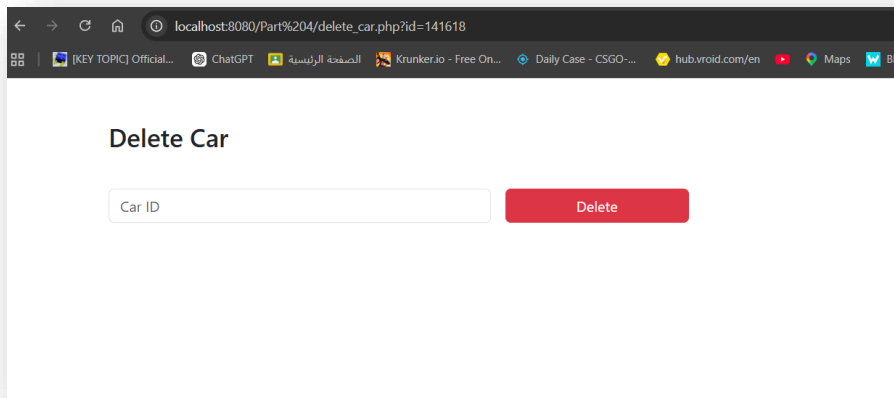
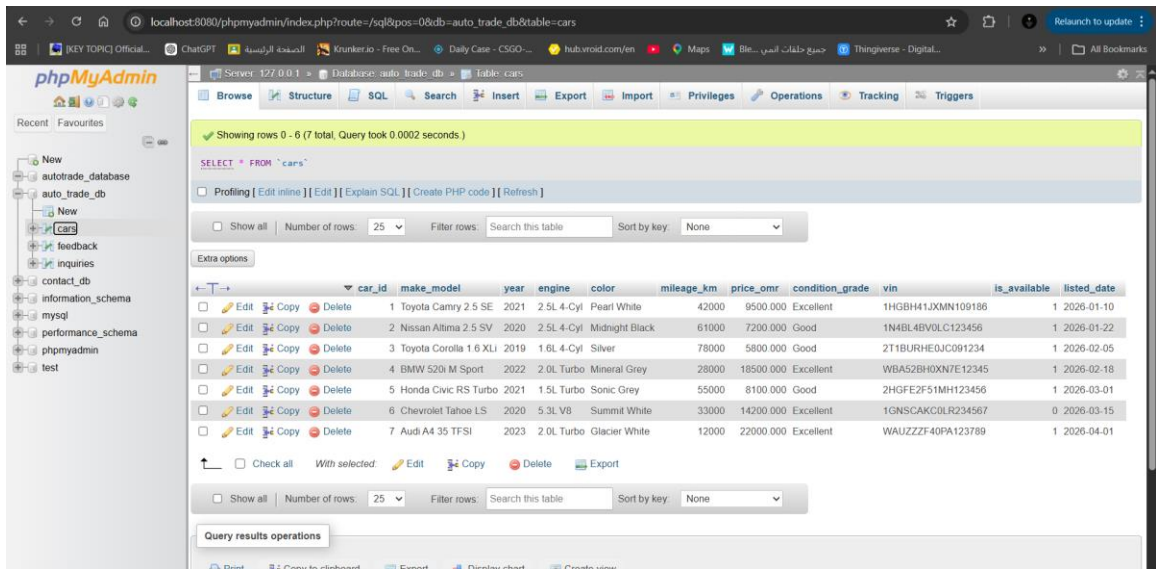
3. User Interface and Experience

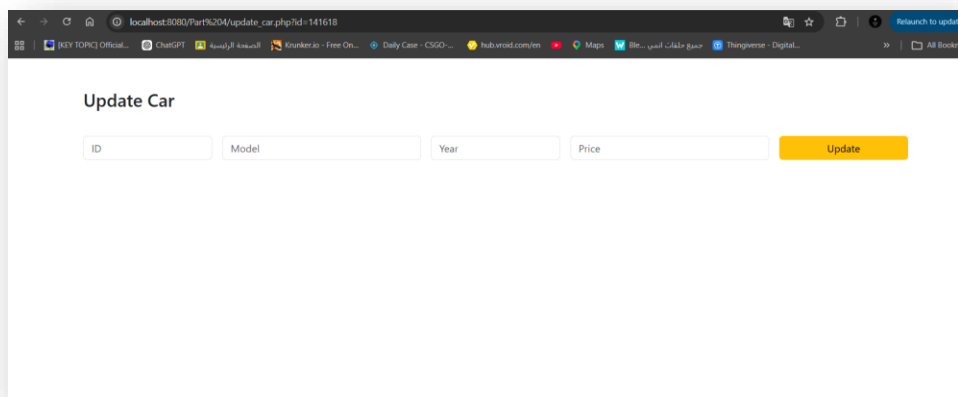
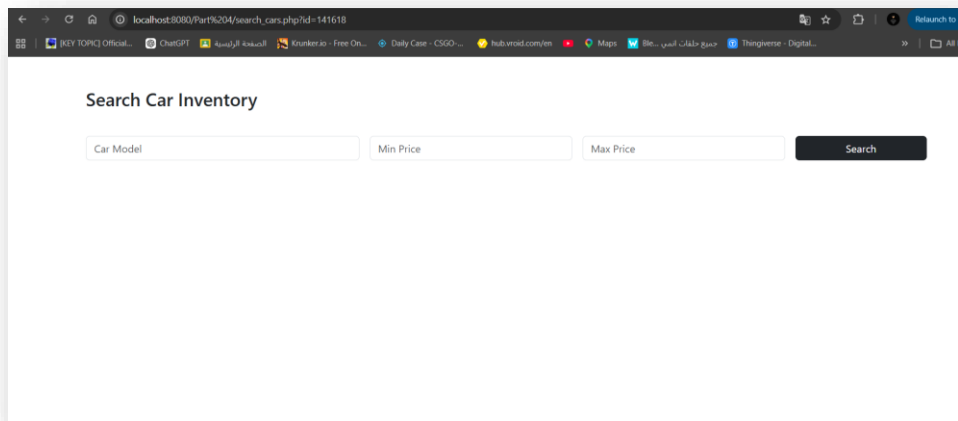
The frontend is designed to be functional and responsive, communicating directly with the backend scripts.

- Sales and Contact Pages: These pages include forms that capture user input and send it to the PHP backend for database insertion.
- Management Dashboard: Provides a clear interface for administrators to view and manage the current car inventory and feedback logs.

Screenshots:







link to the GitHub repository :

https://github.com/s141618-squ/COMP3700_Project2_TheBuilders_SP26/tree/main